AN INDUCTION PRINCIPLE FOR PROPERTIES OF QUASI-COMPACT QUASI-SEPARATED OBJECTS OF THE SMALL ETALE SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a representable-case transfer of the scheme induction principle, a descending induction along the decent-spaces filtration, and a finite chain refinement through affine decompositions of the non-affine legs, all inside the delivered span

Editorial note. The label lemma-induction-principle names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Elementary distinguished squares of algebraic spaces and the induction principle for quasi-compact quasi-separated schemes from Cohomology of Schemes
- The filtration lemma producing the descending chain of opens with affine-at-closing etale maps, from Decent Spaces
- The refinement lemma turning a non-affine leg into a chain of elementary distinguished squares with affine legs

Source reference. Original lemma 08DR: sources/stacks-chow/tags.txt, source lines 10646–10646. Prerequisite pointer invoked at proof line(s) 1595; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07ST: sources/stacks-chow/tags.txt, source lines 9913–9913. Prerequisite pointer invoked at proof line(s) 1608; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08GN: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1541–1541. Prerequisite pointer invoked at proof line(s) 1634; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Let P be a property of the quasi-compact and quasi-separated objects of $X_{spaces,étale}$. Assume that*

(1) *P holds for every affine object of $X_{spaces,étale}$,*

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(2) *for every elementary distinguished square* $(U \subset W, f : V \to W)$ *such that*
   (a) W *is a quasi-compact and quasi-separated object of* $X_{spaces,étale}$,
   (b) U *is quasi-compact,*
   (c) V *is affine, and*
   (d) P *holds for* U, V, *and* $U \times_W V$,
   then P *holds for* W.

Then P *holds for every quasi-compact and quasi-separated object of* $X_{spaces,étale}$ *and in particular for* X.

Proof. We first claim that P holds for every representable quasi-compact and quasi-separated object of $X_{spaces,étale}$. Namely, suppose that $U \to X$ is étale and U is a quasi-compact and quasi-separated scheme. By assumption (1) property P holds for every affine open of U. Moreover, if $W, V \subset U$ are quasi-compact open with V affine and P holds for W, V, and $W \cap V$, then P holds for $W \cup V$ by (2) (as the pair $(W \subset W \cup V, V \to W \cup V)$ is an elementary distinguished square). Thus P holds for U by the induction principle for schemes, see Cohomology of Schemes, Lemma 08DR.

To finish the proof it suffices to prove P holds for X (because we can simply replace X by any quasi-compact and quasi-separated object of $X_{spaces,étale}$ we want to prove the result for). We will use the filtration

$$\emptyset = U_{n+1} \subset U_n \subset U_{n-1} \subset \ldots \subset U_1 = X$$

and the morphisms $f_p : V_p \to U_p$ of Decent Spaces, Lemma 07ST. We will prove that P holds for U_p by descending induction on p. Note that P holds for U_{n+1} by (1) as an empty algebraic space is affine. Assume P holds for U_{p+1}. Note that $(U_{p+1} \subset U_p, f_p : V_p \to U_p)$ is an elementary distinguished square, but (2) may not apply as V_p may not be affine. However, as V_p is a quasi-compact scheme we may choose a finite affine open covering $V_p = V_{p,1} \cup \ldots \cup V_{p,m}$. Set $W_{p,0} = U_{p+1}$ and

$$W_{p,i} = U_{p+1} \cup f_p(V_{p,1} \cup \ldots \cup V_{p,i})$$

for $i = 1, \ldots, m$. These are quasi-compact open subspaces of X. Then we have

$$U_{p+1} = W_{p,0} \subset W_{p,1} \subset \ldots \subset W_{p,m} = U_p$$

and the pairs

$$(W_{p,0} \subset W_{p,1}, f_p|_{V_{p,1}}), (W_{p,1} \subset W_{p,2}, f_p|_{V_{p,2}}), \ldots, (W_{p,m-1} \subset W_{p,m}, f_p|_{V_{p,m}})$$

are elementary distinguished squares by Lemma 08GN. Note that P holds for each $V_{p,1}$ (as affine schemes) and for $W_{p,i} \times_{W_{p,i+1}} V_{p,i+1}$ as this is a quasi-compact open of $V_{p,i+1}$ and hence P holds for it by the first paragraph of this proof. Thus (2) applies to each of these and we inductively conclude P holds for $W_{p,1}, \ldots, W_{p,m} = U_p$. $\square$